

Quantitative Reasoning – Solutions

(Red-Starred Questions)

Q1. Set Theory / Venn Diagrams (Q4)

Given: $n(A)=18$, $n(B)=22$, $n(C)=25$, $n(A \cap B)=8$, $n(A \cap C)=9$,
 $n(B \cap C)=10$, $n(A \cap B \cap C)=5$, total workers = 50.

$$(i) \ n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

$$= 18 + 22 + 25 - 8 - 9 - 10 + 5$$

$$= 43$$

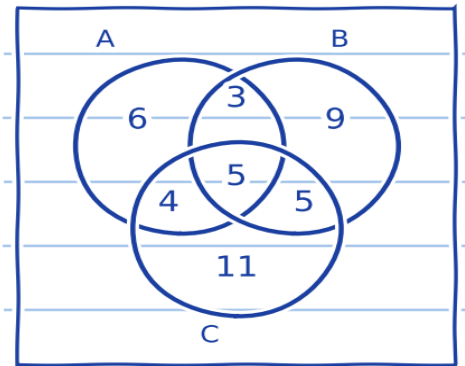
$$(ii) \text{ Not assigned to any zone} = 50 - 43 = 7$$

Region counts (Venn diagram):

$$A \cap B \text{ only} = 8 - 5 = 3; \ A \cap C \text{ only} = 9 - 5 = 4; \ B \cap C \text{ only} = 10 - 5 = 5$$

$$A \text{ only} = 18 - 3 - 4 - 5 = 6; \ B \text{ only} = 22 - 3 - 5 - 5 = 9; \ C \text{ only} = 25 - 4 - 5 - 5 = 11$$

Ans: (i) 43 workers (ii) 7 workers



Q2. Set Theory / Venn Diagrams (Q5)

Given: total = 100, $n(C)=55$, $n(F)=45$, $n(B)=40$,
 $n(C \cap F)=20$, $n(C \cap B)=18$, $n(F \cap B)=15$, $n(C \cap F \cap B)=10$.

$$a) \text{ Exactly two} = (20 - 10) + (18 - 10) + (15 - 10)$$

$$= 10 + 8 + 5 = 23$$

$$b) \text{ Only cricket} = 55 - 20 - 18 + 10 = 27$$

$$c) \text{ Only football} = 45 - 20 - 15 + 10 = 20$$

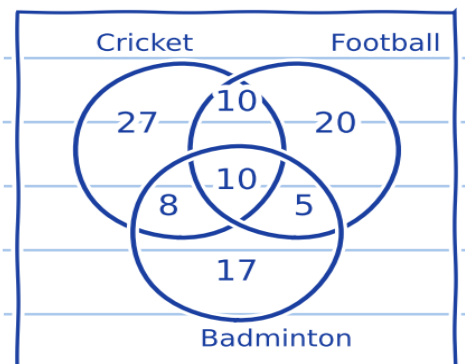
$$(\text{Only badminton} = 40 - 18 - 15 + 10 = 17)$$

$$d) \text{ At least one} = 55 + 45 + 40 - 20 - 18 - 15 + 10 = 97$$

$$e) \text{ None} = 100 - 97 = 3$$

f) Venn diagram regions: only C=27, only F=20, only B=17, C&F only=10, C&B only=8, F&B only=5, all three=10.

Ans: a) 23 b) 27 c) 20 d) 97 e) 3



Q3. Probability (Q3)

Two fair dice are rolled. Total outcomes = 36.

(a) Prime sums: 2, 3, 5, 7, 11 → 1 + 2 + 4 + 6 + 2 = 15 outcomes

$$P = 15/36 = 5/12$$

(b) Sum > 7: 8, 9, 10, 11, 12 → 5 + 4 + 3 + 2 + 1 = 15 outcomes

$$P = 15/36 = 5/12$$

(c) Sum even = 18 outcomes; minus 6 doubles = 12 outcomes

$$P = 12/36 = 1/3$$

(d) Sum = 6: (1,5),(2,4),(3,3),(4,2),(5,1) → 5 outcomes

$$P = 5/36$$

Ans: (a) 5/12 (b) 5/12 (c) 1/3 (d) 5/36

Q4. Probability (Q4)

Total products = 1200; defective = 18 + 30 = 48; non-defective = 1152.

(a) $P(\text{defective}) = 48/1200 = 0.04$

(b) $P(\text{non-defective}) = 1 - 0.04 = 0.96$

(c) Expected defective in 5000 = $0.04 \times 5000 = 200$

(d) Expected non-defective in 5000 = $0.96 \times 5000 = 4800$

Ans: (a) 0.04 (b) 0.96 (c) 200 (d) 4800

Q5. Permutations & Combinations (Q2)

Six students A, B, C, D, E, F stand in a row.

(i) A and B together:

Treat AB as one block → 5 objects → 5! ways.

Within the block: 2! ways (AB or BA).

$$\text{Total} = 5! \times 2! = 120 \times 2 = 240$$

(ii) A and B not together:

$$\text{Total arrangements} = 6! = 720$$

$$\text{Not together} = 720 - 240 = 480$$

Ans: (i) 240 (ii) 480

Q6. Permutations & Combinations (Q4)

COMPUTER has 8 different letters.

(i) Select and arrange 5 letters:

$$P(8,5) = 8!/3! = 8 \times 7 \times 6 \times 5 \times 4 = 6720$$

(ii) Letter C first: fix C in the first place, then arrange 4 of the remaining 7 letters:

$$P(7,4) = 7!/3! = 7 \times 6 \times 5 \times 4 = 840$$

Ans: (i) 6720 (ii) 840

Q7. Area & Perimeter (Q1)

Rectangle $10\text{ m} \times 8\text{ m}$, semicircle (diameter 8 m , $r = 4\text{ m}$) on one shorter side, triangle (base 8 m , height 6 m , equal sides 10 m) on the other.

Area = Rectangle + Semicircle + Triangle

$$= (10 \times 8) + \left(\frac{1}{2} \pi 4^2\right) + \left(\frac{1}{2} \times 8 \times 6\right)$$

$$= 80 + 8\pi + 24 = 104 + 8\pi$$

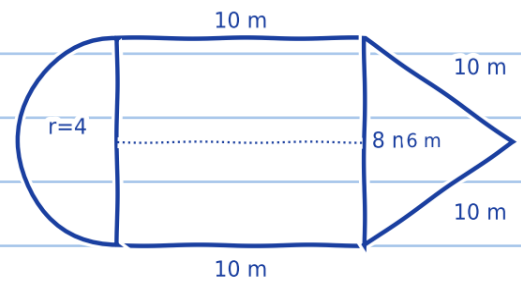
$$\approx 104 + 25.13 = 129.13\text{ m}^2$$

Perimeter = $10 + 10 + (\text{semicircle arc}) + 10 + 10$

$$= 40 + \pi \times 4 = 40 + 4\pi$$

$$= 40 + 12.57 = 52.57\text{ m}$$

Ans: Area $\approx 129.13\text{ m}^2$ Perimeter $\approx 52.57\text{ m}$



Q8. Area & Perimeter (Q2)

Rectangle $18\text{ m} \times 10\text{ m}$ with a semicircle ($r = 5\text{ m}$) on one shorter side.

Area = Rectangle + Semicircle

$$= (18 \times 10) + \left(\frac{1}{2} \pi 5^2\right)$$

$$= 180 + 12.5\pi = 180 + 39.27$$

$$= 219.27\text{ m}^2$$

Perimeter = $18 + 18 + 10 + (\text{semicircle arc})$

$$= 46 + \pi \times 5 = 46 + 5\pi$$

$$= 46 + 15.71 = 61.71\text{ m}$$

Ans: Area $\approx 219.27\text{ m}^2$ Perimeter $\approx 61.71\text{ m}$

